

One Step Reverse Transcription Polymerase Chain Reaction (RT-PCR)

Convert RNA into cDNA, which is then amplified using PCR to produce millions of copies of the target sequence.

Reagents

✓	Name	Quantity	Location/Notes
☐	OneTaq One-Step Reaction Mix (2X)	25 µl	
☐	OneTaq One-Step Enzyme Mix (25X)	2 µl	
☐	Gene-specific Forward Primer (10 µM)	2 µl	
☐	Gene-specific Reverse Primer (10 µM)	2 µl	
☐	Nuclease-free H2O	19-x µl	
☐	Total RNA (up to 1 µg)	x µl	
☐	Hot Start OneTaq DNA polymerase (not provided)	2 units	Neg control

Equipment

✓	Name	Quantity	Location/Notes
☐			
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☐			

Instructions

[One Step RT PCR]

1. Thaw system components and mix by inverting several times.
2. Mix the following components, except RNA, in sterile RNase-free microfuge tubes.
 - a. Note: Do math before reactions

Reagent Tracker	Quantity
☐ OneTaq One-Step Reaction Mix (2X)	25 μ l
☐ OneTaq One-Step Enzyme Mix (25X)	2 μ l
☐ Gene-specific Forward Primer (10 μ M)	2 μ l
☐ Gene-specific Reverse Primer (10 μ M)	2 μ l
☐ Nuclease-free H ₂ O	19-x μ l
☐ Total RNA (up to 1 μ g)	x μ l
Total volume	50 μ l

3. Add RNA template last, and start reactions immediately, as follows with thermocycler:

CYCLE STEP	TEMPERATURE	TIME	CYCLES
Reverse Transcription	48°C	15–30 minutes	1
Initial Denaturation	94°C	1 minute	1
Denaturation	94°C	15 seconds	40
Annealing	50–65°C	30 seconds	
Extension	68°C	1 minute per kb	
Final Extension	68°C	5 minutes	1
Hold	4°C	∞	1

[If you need to denature the template first]

1. Mix RNA sample and primers in sterile, RNase-free microfuge tubes.

Reagent Tracker	Quantity
☐ Total RNA (up to 1 µg)	x µl
☐ Gene-specific reverse primer (10 µM)	2 µl
☐ OneTaq One-Step Reaction Mix (2X)	25 µl
☐ Nuclease-free H ₂ O	19–x µl
Total volume	46 µl

2. Denature for 5 minutes at 65°C. Put promptly on ice.
3. Add the following components to the tube:
 - a. OneTaq One-Step Enzyme Mix (25X) 2 µl
 - b. Gene-specific Forward Primer (10 µM) 2 µl
4. Add tubes to the thermocycler, and run the following program

CYCLE STEP	TEMPERATURE	TIME	CYCLES
Reverse Transcription	48°C	15–30 minutes	1
Initial Denaturation	94°C	1 minute	1
Denaturation	94°C	15 seconds	40
Annealing	50–65°C	30 seconds	
Extension	68°C	1 minute per kb	
Final Extension	68°C	5 minutes	1
Hold	4°C	∞	1

[No-RT Negative Control Reaction]

1. Mix the following components in a sterile RNase-free microfuge tube

2.

Reagent Tracker	Quantity
☐ OneTaq One-Step Reaction Mix (2X)	25 µl
☐ Gene-specific Forward Primer (10 µM)	2 µl
☐ Gene-specific Reverse Primer (10 µM)	2 µl
☐ Hot Start OneTaq DNA polymerase (not provided)	2 units
☐ Nuclease-free H ₂ O	19-x µl
☐ Total RNA (up to 1 µg)	x µl
Total volume	50 µl

3. Proceed with reaction conditions as follows:

CYCLE STEP	TEMPERATURE	TIME	CYCLES
Reverse Transcription	48°C	15–30 minutes	1
Initial Denaturation	94°C	1 minute	1
Denaturation	94°C	15 seconds	40
Annealing	50–65°C	30 seconds	
Extension	68°C	1 minute per kb	
Final Extension	68°C	5 minutes	1
Hold	4°C	∞	1

Notes

Resources:

- https://www.neb.com/en-ca/protocols/one-step-rt-pcr-protocols-e5315?srsId=AfmBOopjMeOMNIsP_WbnhZhKcu6zamt38Mzxuo0mdbBfM29PpnwqbB6v

References